CPTG Primordial Light-Element Abundances

A Frozen-Output Observational Addendum to the Complete-Coordinate PRIMAT Validation

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Abstract

We compare the previously frozen Curvature Polarization Transport Gravity (CPTG) primordial helium-4 mass fraction, $Y_{\mathrm{p}}^{\mathrm{CPTG}} = 0.24576024693340703$, with two independently published 2026 observational determinations. Relative to the original observational benchmark, which placed CPTG at $+0.253416\sigma$, the updated Large Binocular Telescope (LBT) measurement, $Y_{\mathrm{p}} = 0.2458 \pm 0.0013$, moves the discrepancy substantially closer to the observational center, yielding just -0.030579σ. The absolute observation-normalized residual therefore falls from 0.253416σ to 0.030579σ, an 87.93% reduction (a factor of 8.29), placing the unchanged CPTG result in close agreement with the LBT central value. The independent EMPRESS XV determination, $Y_{\mathrm{p}} = 0.2402 \pm 0.0040$, yields $+1.390062\sigma$, lying outside the quoted 1σ interval but within 2σ. For both 2026 determinations, the frozen CPTG abundance lies closer to the respective observational center than the archived same-baryon standard-radiation calibration. No CPTG equations, parameters, or calculated abundances were modified. The improved LBT agreement therefore arises entirely from the updated observational comparison, not from theoretical adjustment. This addendum documents the new observational results, their publication chronology, and a self-contained reproduction package.

1 Introduction

In *The Cosmological Lithium Problem and the CPTG Lithium Solution* [1], the fixed CPTG primordial comparison coordinates and mass-seven transport law were implemented in the native PRIMAT reaction network. The study reported a reduction in the computed ^{7}Li/H abundance from $5.2735647904 \times 10^{-10}$ to $1.6789263563 \times 10^{-10}$, corresponding to $+0.915705\sigma$ against its declared lithium observational anchor, while leaving deuterium and helium abundances essentially unchanged by the mass-seven transport gate. All 23 preregistered scientific validation gates passed. The helium-4 abundance and same-baryon standard-radiation calibration archived in that study provide the frozen numerical basis for this addendum; the earlier network validation is not repeated.

The purpose of this addendum is narrower: to place those *unchanged numerical results* beside the updated 2026 Large Binocular Telescope (LBT) and EMPRESS XV observational helium determinations, using each study's own published central value and uncertainty. Both surveys are retained, regardless of their differing central estimates. The addendum records how the observational sigma residual changes when the observational reference changes, and separately shows the effect of the previously archived radiation-coordinate control. No physical coefficient, radiation coordinate, or calculated abundance is updated in response to the measurements.

2 Previously Frozen Calculation and Control

The CPTG source calculation used the registered primordial baryon and radiation comparison coordinates $\eta_{10,\mathrm{BBN}} = 5.998071834744$ and $N_{\mathrm{eff}}^{\mathrm{CPTG}} = 2.968584073464$, as reported in the parent paper [2, 1]. Only the resulting published helium outputs enter the present analysis; the geometric transport derivation and the original PRIMAT execution are not repeated.

The parent study also retained a *same-baryon standard-radiation calibration* before the fixed CPTG radiation-target calculation. Both are *previously archived numerical states*, not observational determinations and not alternative fits made for LBT or EMPRESS. Their source provenance and earlier authority identifiers are summarized in Appendix A.

3 Observational Helium Data and Publication Chronology

3.1 LBT: high-precision metal-poor H II-region analysis

Aver et al. reported a primordial ^{4}He mass fraction from Large Binocular Telescope observations of 54 metal-poor H II regions [4]. After quality screening and selection of 15 high-signal-to-noise, low-metallicity targets for the adopted weighted-mean estimator, they obtained

$$Y_{\mathrm{p}}^{\mathrm{LBT}} = 0.2458 \pm 0.0013. \tag{1}$$

The analysis explicitly considers observational systematics and updated helium emissivities. Its January 29, 2026 arXiv preprint precedes the August CPTG network lock; the journal article appeared September 10, 2026. Therefore a comparison undertaken in October 2026 is a new *reporting exercise* against an already published observation, not an automatically prospective prediction of data unavailable during the original CPTG calculation.

3.2 EMPRESS XV: independent galaxy-spectroscopy determination

Yanagisawa et al.'s EMPRESS XV study analyzes an extremely metal-poor-galaxy sample using optical and near-infrared spectral information and a photoionization-based helium abundance analysis [5]. The final published journal value is

$$Y_{\mathrm{p}}^{\mathrm{EMPRESS}} = 0.2402 \pm 0.0040. \tag{2}$$

This value is lower than LBT's central estimate and has a wider quoted uncertainty. The earliest EMPRESS XV preprint, dated June 30, 2025, reported a different interim abundance estimate; the present comparison *uses the 2026 published journal value*, not a mixture of manuscript versions. The 2026 journal publication also predates the CPTG August lock. Both observational determinations are reported even though their central values differ.

3.3 Original CPTG controls and reference separation

The source lithium paper declared a helium comparison anchor of $Y_{\mathrm{p}} = 0.245 \pm 0.003$ [1]. We retain that historical anchor solely to distinguish the original reported comparison from the LBT and EMPRESS determinations. The three observational references are separate measurement inputs; no new CPTG helium calculation is performed.

Table 1 preserves the relevant order of public availability and numerical execution.

Date	**Event**	**Role**
30 Jun 2025	EMPRESS XV initial public preprint (later revised)	Observed
29 Jan 2026	LBT helium measurement released as arXiv preprint	Observed
10 Jun 2026	EMPRESS XV final journal publication	Observed
19 Aug 2026	CPTG complete-coordinate PRIMAT final evidence and authority lock	Frozen
10 Sep 2026	LBT paper published in *The Astrophysical Journal*	Observed
23 Oct 2026	Present numerical cross-check and draft addendum	Report

Table 1: Public-observation and CPTG validation chronology. The journal publication date is not substituted for an earlier preprint release date when assessing whether a test is strictly prospective.

4 Comparison Methodology

4.1 One fixed theory output, multiple observational references

The helium value used throughout is the previously frozen gate-off PRIMAT result [1]:

$$Y_{\mathrm{p}}^{\mathrm{CPTG}} = 0.24576024693340703. \tag{3}$$

For an observational determination i with reported central value y_i and quoted standard uncertainty s_i, define its signed, observation-normalized residual as

$$z_i = \frac{Y_{\mathrm{p}}^{\mathrm{CPTG}} - y_i}{s_i}. \tag{4}$$

Positive z_i means the fixed CPTG abundance exceeds the observational central value. Negative z_i means it falls below. This is a transparent normalization by the measurement's reported uncertainty; it is *not* a combined uncertainty including a separately established theoretical error budget, nor a joint likelihood across surveys. No y_i, s_i, or z_i enters the theoretical calculation.

To isolate the already archived radiation-coordinate change, Equation (4) is also evaluated using the standard-radiation calibration abundance

$$Y_{\mathrm{p}}^{\mathrm{cal}} = 0.2467730506356945, \tag{5}$$

which shares CPTG's baryon coordinate and the gate-off native network, but uses the code-local $\Delta N_{\mathrm{eff}} = 0$ calibration background. These are two previously frozen calculated states. The original helium anchor and the 2026 LBT and EMPRESS values are observational references; neither calculated state was updated for this addendum.

5 Helium Comparison Results

5.1 Absolute observational residuals

Table 2 reports the comparison using both the preserved standard-radiation calibration and CPTG's fixed radiation target. It includes the original paper's reference, the LBT 2026 result, and the EMPRESS XV published result.

Helium reference	**Observed Y_{p}**	**Calibration z**	**CPTG z**
Original source anchor	0.245 ± 0.003	$+0.591017\sigma$	$+0.253416\sigma$
LBT 2026	0.2458 ± 0.0013	$+0.748500\sigma$	$\mathbf{-0.030579\sigma}$
EMPRESS XV 2026	0.2402 ± 0.0040	$+1.643263\sigma$	$+1.390062\sigma$

Table 2: Signed helium-4 residuals from two previously archived calculated states, normalized to each observational reference: the same-baryon standard-radiation calibration and the frozen CPTG radiation-target output. The rows are observational sources, not separate theoretical fits. Neither calculated value is updated in this addendum.

For LBT, using the full frozen helium value,

$$z_{\mathrm{LBT}} = \frac{0.24576024693340703 - 0.2458}{0.0013} \approx -0.030579282\sigma. \tag{6}$$

For EMPRESS XV the corresponding arithmetic is

$$z_{\mathrm{EMPRESS}} = \frac{0.24576024693340703 - 0.2402}{0.0040} \approx +1.390061733\sigma. \tag{7}$$

The same unchanged CPTG output falls about 0.031 observational standard deviations below the LBT central value and about 1.39 above the EMPRESS XV central value. Figure 1 shows only this *one frozen CPTG theoretical value* alongside the original observational anchor and the two 2026 measurements. The archived calibration is shown in Table 2 for provenance, not as a second theoretical prediction in the figure. The LBT and EMPRESS central values differ by 0.0056 in Y_{p}; treating their quoted errors as statistically independent for this descriptive calculation gives about 1.33 combined standard deviations, without accounting for possible shared systematics.

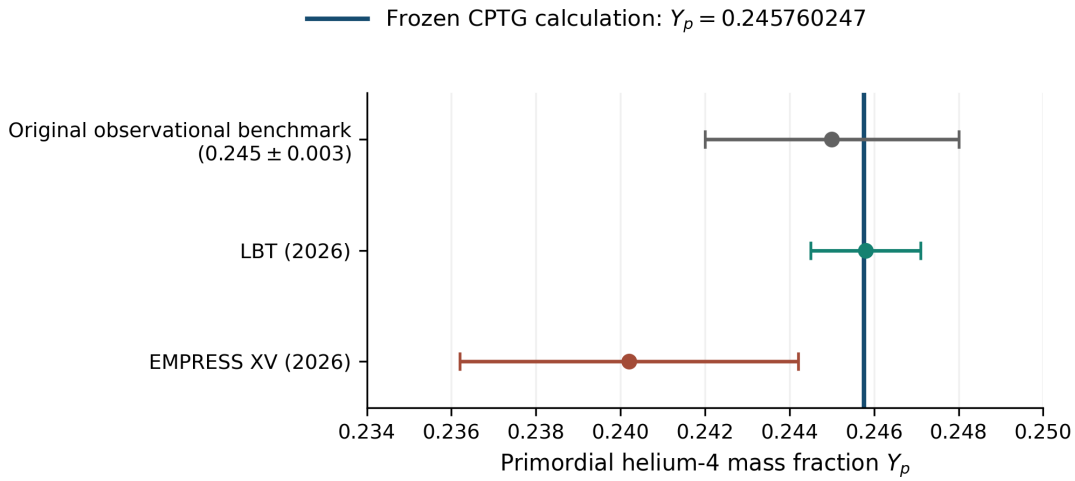


Figure 1: The single, previously frozen CPTG helium mass fraction (blue vertical line, labeled "Frozen CPTG calculation") and three observational helium determinations (points with reported one-standard-deviation intervals). The original observational benchmark (0.245 ± 0.003) is shown for historical context; the LBT and EMPRESS measurements are the updated observational comparisons. No CPTG output has been recalculated.

5.2 Signed residuals against the same frozen CPTG value

Figure 2 expresses each observational comparison on its own quoted one-standard-deviation scale using Equation (4). The original observational benchmark yields $+0.253416\sigma$; the updated LBT and EMPRESS XV references yield -0.030579σ and $+1.390062\sigma$, respectively. The changing position of the points reflects *different observational central values and uncertainties*, not any change to the CPTG calculation. The historical standard-radiation calibration remains separately listed in Table 2.

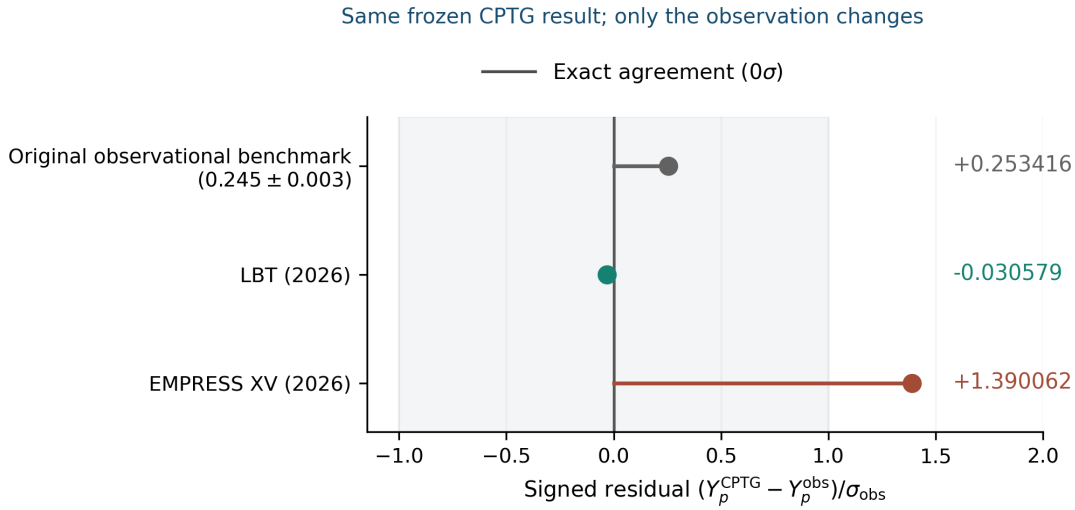


Figure 2: Signed observational residuals of the same frozen CPTG helium value against the original observational benchmark (0.245 ± 0.003) and the published 2026 LBT and EMPRESS XV determinations. The vertical line labeled "Exact agreement (0σ)" denotes equality with an observational central value; the pale band spans -1σ to $+1\sigma$. Each dot represents a different observational reference, not a newly fitted or updated CPTG prediction.

6 Interpretation and Limits

The LBT measurement is particularly close to the archived CPTG central value, while the published EMPRESS XV value provides a less close, but still separately documented, comparison. The same frozen CPTG abundance is used in every row. The observed differences among the surveys are not used to choose a new CPTG physical coordinate or a new helium abundance.

The archived calibration row is retained to identify the effect of CPTG's *already fixed* radiation-coordinate translation at one baryon coordinate. It is not a measurement, an old observational baseline, or a general substitute for an independently specified competing physical model. The smaller absolute residuals obtained from the frozen CPTG row are conditional on these particular observational central values and quoted uncertainties.

The residuals in this paper divide by observational uncertainties only; they do not constitute a combined theoretical error model or a joint likelihood. The different survey reductions may also have systematic uncertainties not represented by a simple independent-Gaussian comparison. Both helium determinations are therefore reported separately. Because the relevant observational results were public before the cited August 2026 CPTG authority

lock, the arithmetic presented here does not by itself establish a strictly pre-data prediction. The already published reaction-network and mass-seven tests remain upstream evidence and are not repeated or regraded in this observational addendum.

7 Conclusions

The previously frozen CPTG helium abundance $Y_{\mathrm{p}} = 0.24576024693340703$ gives observation-normalized residuals of -0.030579σ against the 2026 LBT result and $+1.390062\sigma$ against the published EMPRESS XV result. Relative to the separately archived same-baryon standard-radiation calibration, the CPTG radiation-coordinate state is closer to each of these observational central values. These are comparisons of preserved calculations with observational measurements, not new theoretical outputs.

The addendum supplies source provenance and a reproducible record of the observational arithmetic while retaining the original CPTG numerical authority unchanged. The earlier multi-species nuclear-network and mass-seven validation remain in the cited parent paper rather than being re-litigated here.

Data, Code, and Evidence Availability

All materials required to reproduce the *new observational comparisons in this addendum* are distributed together in:

`CPTG_2026_Helium_Observational_Addendum_Evidence.zip`

The single archive contains the LaTeX manuscript, rendered review PDF, both publication-resolution PNG figures, the figure-generation source, immutable source-value and observation registries (CSV), the comparison script, its derived results, a provenance note, a README with reproduction steps, and an internal SHA-256 file manifest. The script reproduces the observational residual arithmetic; no nuclear-network computation is required.

The archive is a *new addendum evidence package*, distinct from the existing CPTG PRIMAT authority archives and their published SHA-256 identities (Appendix A). It does not purport to bundle all dependencies needed to independently rerun PRIMAT. Those upstream authority records remain cited in the original lithium paper and its public evidence repository [1, 6].

A Previously Published Authority and Source Identifiers

This appendix identifies the upstream PRIMAT authority without reproducing its scientific gate-by-gate discussion. The parent CPTG lithium study [1] describes the four native PRIMAT runs (two code-local radiation calibration solves and two CPTG target solves with the mass-seven gate off/on), their frozen input specification, and the original validation evidence. The present observational addendum reads the recorded helium values only. The original mass-seven and cross-species controls are neither recalculated nor submitted as new tests here.

Archive or object	**Archived identity reported by CPTG lithium study**
Authority lock	`CPTG_PRIMAT_A7_FullRadiationCoordinateValidation_AuthorityLock_20260819_2359EDT_r01.zip`
Authority-lock SHA-256	`43b65036e229dc5f04d445f5890399236aaf546c598f1711e572043db6b05bec`
Final-evidence archive	`CPTG_PRIMAT_A7_FullRadiationCoordinateValidation_20260819_2344EDT_r05_FINAL_EVIDENCE.zip`
Final-evidence SHA-256	`9c44543cd2497e72170c0a64173423c9b3bf2f1f3924c9d4e77c16a318fb8b4c`
Preregistration SHA-256	`ab38b8c5bdb435b6c311e9b5e0fe31bca1c06ec67d2dc3e4a92098f255425162`
Radiation-mapping SHA-256	`f4e1b0b4e4df47914da9f904d544565ac61a5687879c536560ec8c004b7cbe16`
Final scientific-result SHA-256	`34c0dfdf44d5917ba44d2253c59195dfe258ae4d4de1adf3db6870eafc993626`

Table 3: Upstream frozen authority identifiers cited from the earlier CPTG lithium paper. These digests are published reference identifiers and not hashes of the separate addendum archive.

The previously archived gate-off helium result is 0.24576024693340703 and the previously archived same-baryon standard-radiation calibration result is 0.2467730506356945. Both are treated as immutable source inputs for the new helium comparisons. The addition of an updated observational reference changes a calculated residual, not either source value.

References

[1] C. L. Glass Jr., *The Cosmological Lithium Problem and the CPTG Lithium Solution*, Zenodo (2026). DOI: 10.5281/zenodo.22420032.

[2] C. L. Glass Jr., *CPTG Geometric Pi Branch: Cosmological Comparison Coordinates*, Version II, Zenodo (2026). DOI: 10.5281/zenodo.22907311.

[3] C. L. Glass Jr., *Curvature Polarization Transport Gravity: A Unified Geometric Framework for Structure, Cosmology, and Nuclear Reactions*, version 3, Zenodo (2026). DOI: 10.5281/zenodo.23288198.

[4] E. Aver, E. D. Skillman, R. W. Pogge, N. S. J. Rogers, M. K. Weller, K. A. Olive, D. A. Berg, J. J. Salzer, J. H. Miller and J. E. Méndez-Delgado, *The LBT $Y_{\rm p}$ Project. IV. A New Value of the Primordial Helium Abundance*, *Astrophysical Journal* **1008**, 240 (2026). DOI: 10.3847/1538-4357/ae879b; preprint arXiv:2601.22238 (29 January 2026).

[5] H. Yanagisawa et al., *EMPRESS. XV. A New Determination of the Primordial Helium Abundance Suggesting a Moderately Low $Y_{\rm P}$ Value*, *Astrophysical Journal* **1004**, 55 (2026). DOI: 10.3847/1538-4357/ae64fc; preprint arXiv:2506.24050 (initial posting 30 June 2025; the comparison uses the final 2026 journal value).

[6] CPTG, *Supporting source code, freeze records, numerical outputs, and evidence packages for Curvature Polarization Transport Gravity*, repository mirror (2026). `https://github.com/CLG2025/CPTG`.